

CMake

CMake is a widely-used, cross-platform build system generator with excellent support for mingw-w64. See the [official documentation](#) for more information.

Detect mingw-w64 at Configuration Time

While preprocessor definitions can be used to detect mingw-w64 at compile time, CMake provides variables to detect it during configuration:

```
# CMakeLists.txt
if(MINGW)
    message(STATUS "Building with mingw-w64")
else()
    message(STATUS "Not building with mingw-w64")
endif()
```

Cross compiling on Linux

To cross compile a CMake project for Windows while working on Linux:

- Install a Windows cross compiler on your Linux system
- Create or use a toolchain file that defines how CMake should use the cross compiler
- Configure your project with the toolchain file

For detailed instructions, consult the [official CMake documentation](#).

```
# toolchain-mingw64.cmake
set(CMAKE_SYSTEM_NAME Windows)
set(CMAKE_SYSTEM_PROCESSOR x86_64)

# specify the cross compiler
set(CMAKE_C_COMPILER x86_64-w64-mingw32-gcc)
set(CMAKE_CXX_COMPILER x86_64-w64-mingw32-g++)
set(CMAKE_RC_COMPILER x86_64-w64-mingw32-windres)

# where is the target environment
set(CMAKE_FIND_ROOT_PATH /usr/x86_64-w64-mingw32)
```

```
# search for programs in the build host directories
set(CMAKE_FIND_ROOT_PATH_MODE_PROGRAM NEVER)
# for libraries and headers in the target directories
set(CMAKE_FIND_ROOT_PATH_MODE_LIBRARY ONLY)
set(CMAKE_FIND_ROOT_PATH_MODE_INCLUDE ONLY)

# CMake determines how to examine dependencies based on the *host* system,
leading to
# a `file unknown error` unless the target platform is explicitly specified.
set(CMAKE_GET_RUNTIME_DEPENDENCIES_PLATFORM "windows+pe")
```

```
$ cmake -B build -DCMAKE_TOOLCHAIN_FILE=toolchain-mingw64.cmake
$ cmake --build build
$ wine build/hello.exe
Hello, Windows!
```

